

Extended Euler congruence

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Abstract. The matricial Euler congruence

$$\mathrm{Tr}(A^{p^n}) \equiv \mathrm{Tr}(A^{p^{n-1}}) \pmod{p^n},$$

previously announced in Arnold, *Japanese J. Math.* **1**(1), 1–24, 2006, for $A \in M_N(\mathbb{Z})$, is given a proof based on extending it to the ring of Witt vectors of length n .

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The matricial Euler congruence

$$\mathrm{Tr}(A^{p^n}) \equiv \mathrm{Tr}(A^{p^{n-1}}) \pmod{p^n}, \quad (1)$$

which was announced in [1] for $A \in M_N(\mathbb{Z})$, can be extended as follows. We replace $\mathbb{Z}$ with $\mathbb{Z}/p^n\mathbb{Z}$, and consider $\mathbb{Z}/p^n\mathbb{Z}$ as being $W_n(\mathbb{F}_p)$, the ring of Witt vectors of length n . As geometers, we dislike considering the case of the field $\mathbb{F}_p$ alone: the main part of the following proof depends on the discovery of an extension to the case of $W_n(R)$, where R is of characteristic p .

Proposition. *Let R be an $\mathbb{F}_p$ -algebra, commutative with unity. The endomorphism*

$$F : x \longmapsto x^p$$

of R defines by functoriality an endomorphism F of $W_n(R)$. For A in $M_N(W_n(R))$, the equality

$$\mathrm{Tr}(A^{p^n}) = F(\mathrm{Tr}(A^{p^{n-1}})) \quad (2)$$

holds in $W_n(R)$.

Proof. It suffices to consider the universal case, where R is the ring of polynomials in N^2n variables $X_j^i(k)$ over $\mathbb{F}_p$, and A_j^i is the Witt vector whose components are $X_j^i(k)$. We must prove a polynomial identity in these variables. It suffices to prove it in $W_n(R(1/D))$, where D is the discriminant of the characteristic polynomial of $(X_j^i(0))$. If k is an algebraically closed field containing $R[1/D]$, it suffices to prove (2) in $W_n(k)$: we may and shall assume that $R = k$ is algebraically closed and that the image of A in $M_N(k)$ is a matrix with distinct eigenvalues.

To simplify, we may also suppose that the image of A in $M_N(k)$ is invertible. In this case, the matrix A is conjugate to a diagonal matrix. It remains to prove that, for $a_i \in W_n(k)^*$, $1 \leq i \leq n$, we have

$$\sum_i a_i^{p^n} = F\left(\sum_i a_i^{p^{n-1}}\right) \left(= \sum_i F(a_i^{p^{n-1}})\right).$$

As F is an endomorphism, the task is thus reduced to the case $N = 1$.

It remains to show that, for $a \in W_n(k)$, the element $a^{p^{n-1}}$ is a multiplicative representative, that is, a Witt vector of the form $(\lambda, 0, 0, \dots, 0)$. Indeed, F acts on a multiplicative representative by raising it to the p -th power.

Writing $a = \alpha u$, where α is a multiplicative representative and $u \mapsto 1$ in k , we must prove that $u^{p^{n-1}} = 1$.

The ring $W(k)$ of Witt vectors is a complete discrete valuation ring with residue field k and maximal ideal p , and $W_n(k)$ is its quotient by p^n . We have

$$u \equiv 1 \pmod{p},$$

whence $u^{p^a} \equiv 1 \pmod{p^{a+1}}$ follows inductively, as can be seen from the expansion of $(1 + p^{a-1}x)^p$.

Finally, (1) follows from (2), because F is the identity of $\mathbb{F}_p$. □

References

- [1] V. I. Arnold, *On the matricial version of Fermat–Euler congruences*, *Japanese J. Math.* **1**(1), 1–24 (2006). Publisher’s erratum **1**(2), 469.